Sagar Varma

Computer Engineering student @ UCSB building AI agents, dev infra, and real-time systems

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EXPERIENCE

SemiAnalysis Oct 2025 – Sep 2026

Machine Learning Engineering Intern, Remote

- Built the multi-agent deep research system behind Fab Map, mapping thousands of fabs and cutting research from days to minutes
- Created a ground-truth dataset and LLM evals harness (LLM-as-a-judge) to select models and catch regressions
- Designed verifier agents that catch hallucinations, raising dataset accuracy from ~40% to ~75%
- Developed internal tools for researchers to query and visualize fab data

Avora AI (Y Combinator) Jun 2025 – Sep 2025

Software Engineering Intern, San Francisco, CA

- Scaled the FastAPI backend on Kubernetes to hundreds of dental practices, cutting latency from seconds to milliseconds
- Built an AssemblyAI speech-to-text + LLM pipeline turning appointment audio into clinical notes, summaries and perio charts
- Shipped practice telemetry and quality-of-life features that raised retention 55%

Intel Jun 2022 – Sep 2022

Software Engineering Intern, Folsom, CA

- Built a Python/Perl tool automating code updates across millions of lines, cutting merge conflicts 99%
- Shipped security patches in C for legacy graphics drivers

PROJECTS

Orbit | orbitdeploy.dev | github.com/sagarvarma1/orbit

Open-source, self-hostable Vercel alternative that hosts my projects (TypeScript, Next.js, Docker, Caddy)

- Built Git push-to-deploy CI/CD with preview deployments, Next.js SSR, serverless functions and auto-HTTPS
- Made rollbacks instant with immutable builds and production as a single pointer, so nothing rebuilds
- Sandboxed untrusted code with per-project Linux users, an iptables firewall and Node's permission model; 87-check E2E test suite

Leaf | leaf.orbitdeploy.dev | github.com/sagarvarma1/leaf

Real-time collaborative workspace, Notion-style (TypeScript, WebSockets, ProseMirror, SQLite)

- Wrote a sync engine from scratch (RGA CRDT + Peritext) with offline-first editing via IndexedDB; deployed on Orbit
- Verified convergence with a randomized 3-client simulation across 1,000 seeds, plus 123 unit and end-to-end tests
- Built an AI agent (Claude, tool calling) that edits live as a collaborator; undo reverts only its own changes

Bardie | textbardie.com

Proactive AI assistant over iMessage

- Built an LLM agent with tool calling, Google Calendar/Gmail, vision, and voice input that texts users before they forget
- Launched to 10 early-access users

Appetite Pal | iOS App Store

GLP-1 food log (SwiftUI, iOS)

- Shipped an iOS app with AI calorie and macro tracking from photos and text
- Live on the App Store with subscriptions at $14.99/month or $169.99/year

Earlier projects: americantimeuse.com, wherely.xyz (~1,500 MAU), aitestprep.academy (~1,000 waitlist), wheretowatch.fun

TECHNICAL SKILLS

AI Engineering: LLMs, agentic AI, multi-agent systems, tool calling, MCP, LLM evals, LLM-as-a-judge, RAG, prompt engineering
Machine Learning: PyTorch, TensorFlow, fine-tuning, computer vision, natural language processing (NLP)
Frameworks: Claude Agent SDK, OpenAI Agents SDK, LangGraph, LangChain, LlamaIndex, Claude and OpenAI APIs, Claude Code
Languages: Python, TypeScript, JavaScript, Swift, Java, C, C++, SQL
Web & Mobile: FastAPI, Django, Flask, Node.js, React (React.js), Remix, SQLAlchemy, Pydantic, REST APIs, OAuth 2.0, SwiftUI (iOS)
Infrastructure: PostgreSQL, Redis, vector databases, BigQuery, Docker, Kubernetes, AWS, GCP, GitHub Actions (CI/CD)
Hardware: Verilog, VLSI design, Cadence Virtuoso, Cadence SKILL, electronic design automation (EDA), computer architecture

EDUCATION

University of California, Santa Barbara (UCSB) Expected Jun 2027

Bachelor of Science (B.S.), Computer Engineering

- Coursework: AI in Chip Design, VLSI Design, Computer Architecture, Compilers, Semiconductor Fabrication Technology

ADDITIONAL EXPERIENCE

Independent AI Consulting, agents and internal tools for companies (launching) 2026 – Present

Perplexity, Campus Growth Ambassador at UCSB; drove ~500 signups Sep 2025 – Dec 2025